

Annie Cai

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Education

University of Waterloo

Waterloo, ON

B.ASc in Systems Design Engineering - GPA: 3.99/4.00, 94%

2022 - 2026

- *Awards:* 5x Term Dean's List, President's Undergraduate Research Award, Faculty of Engineering Scholarship, President's Distinction Scholarship
- *Relevant Courses:* Data Structures, Algorithms, Databases, OS, Networks, Distributed Systems, AI/ML

Skills

Languages: Python, Go, C++, Java, Ruby, TypeScript, JavaScript, SQL, GraphQL, HTML/CSS

Technologies: Apache Spark, Apache Kafka, React, Flask, Node.js, Next.js, Express.js, Django

Tools: Git, AWS, PostgreSQL, MySQL, MongoDB, Kubernetes, Docker, Redis, Envoy, Jenkins, Terraform

Work Experience

Stripe 🔗

New York, NY

Software Engineer Intern

May. 2026 - Aug. 2026

- Building ML models and evaluation services to predict payment network costs, optimizing Stripe's largest expense

Snowflake 🔗

Bellevue, WA

Software Engineer Intern

Sep. 2025 - Dec. 2025

- Designed high-throughput traffic simulations for **Envoy ingress** to benchmark and generate autoscaling metrics
- Built a **distributed load-testing pipeline** in **Go** and **JMeter** to detect ingress performance regressions
- Improved traffic-layer reliability by identifying autoscaling bottlenecks and **shipping postmortem-driven** Envoy config changes

Datadog 🔗

New York, NY

Software Engineer Intern

Jan. 2025 - Apr. 2025

- Engineered and deployed a **Kubernetes-based Kafka streaming service** processing 100k+ events/sec for real-time database monitoring
- Parallelized Go event-processing pipelines with worker pools and batching, **increasing throughput by 2x**
- Designed a **distributed caching layer** to speed up API responses and reduce database query latency

Newfront (YC W18) 🔗

San Francisco, CA

Software Engineer Intern

May. 2024 - Aug. 2024

- Developed a scalable **Websockets** pub/sub message system for real-time AI chatbot communication
- Improved **MySQL** database read/write speeds through strategic indexing and schema optimizations

Royal Bank of Canada 🔗

Toronto, ON

Software Engineer Intern

Sep. 2023 - Dec. 2023

- Streamlined ETL by modularizing **Python** pipeline, enabling parallel execution and enhancing workflow efficiency
- Achieved a **40% cost reduction** in storage expenses by **migrating 8GB+** of data to **AWS S3 buckets**
- Scaled database retrieval job during high load periods using **Redis** and **Apache Spark**, leveraging resource tuning and caching strategies to **lower job time from hours to under 5 minutes**

Extracurriculars

Undergraduate Research C++

Developed and evaluated crash-resilient synchronization primitives for persistent memory. Prototyped a recoverable MCS-style queue lock in C++/PMDK, enabling fault-tolerant coordination.

UW Blueprint 🔗 React, TypeScript, Express.js, PostgreSQL, Docker, Heroku

Directed a cross-functional student team to ship a full-stack pet management platform for a nonprofit.